PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville expected 2027
Ph.D. in Economics
Committee: James Lake (Chair), Georg Schaur, Matthew Van Essen, Timothy Fitzgerald

Shandong University, Jinan 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019)

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

International Trade, Trade Policy, Environmental and Energy Economics, and Applied Econometrics.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports [JMP]. [paper]
Abstract: This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm's US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity responses are not those of a markup cut on an unchanged good, and the decline is not the input-tariff upgrading of the same years, which varies only within firm and year and is absorbed by the design. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake.
Abstract: A Trade Restrictiveness Index (TRI) is the uniform tariff that is welfare-equivalent to a country's actual tariff structure. Its standard implementation assigns a single tariff to each product, yet preferential trade agreements make tariffs on the same product differ across exporters, distorting where a country buys as well as what it buys. In an Armington model, we derive a TRI that separates these two distortions: for a small country, the squared index is, to second order, the square of the conventional index evaluated at mean product tariffs plus a term in the variance of tariffs across each product's exporters. We also derive the index for large importers, whose tariffs affect world prices. Across 123 importers over 2000–2020, the dispersion term accounts for 24 percent of the median importer's squared TRI in 2018, up from zero in the early 2000s. Preferences therefore reduce the TRI far less than they reduce average tariffs: in 2018 they cut the median importer's simple average tariff by 16 percent but its TRI by only 5 percent.

Estimating Mean-Field Games with a Payoff Threshold: Candidates and Party Lists in

Brazil with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*

Abstract: Estimating a discrete-choice game of incomplete information requires evaluating every joint action profile, which is infeasible when dozens of candidates interact, as in Brazilian council elections. We instead adopt a mean-field equilibrium, in which candidates affect one another only through list shares, reducing the fixed point to one dimension per list. Because seats go only to lists that clear a threshold, an equilibrium exists if and only if some proposed set of clearing lists is self-fulfilling, and we show that such a set need not exist. Since several lists can clear the threshold in the same market, within-market log-odds contrasts difference out unobserved list appeal and identify how responsiveness differs across candidate types. In markets of a few dozen candidates, sampling noise in list shares attenuates the maximum-likelihood estimate of the weight on the chance of a seat; a split-sample instrument removes most of this bias in simulations. Across 926,900 council candidacies, candidates favor lists with a better chance of a seat, and among non-incumbents the response is stronger for those with a university degree.

PEER-REVIEWED PUBLICATIONS

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech — Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series — Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group — Spring 2024, Fall 2024
Western Regional Science Association, Annual Meeting, Hawaii — 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM — 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA — 2022

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics — Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development — Summer 2025
ECON 313: Intermediate Macroeconomics — Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024
Master's courses:	
AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Tennessee	2021 – present
Graduate Research Assistant, Shandong University	2016 – 2018

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee	2026
Summer Graduate Assistantship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Academic Scholarship for Master's Students, Shandong University	2017
Freshman Scholarship, Shandong University	2016
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015
National Endeavor Scholarship, Tongji University	2014
Tongji Scholarship of Excellence, Third Prize, Tongji University	2014

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Languages: Chinese (native), English (fluent).

REFERENCES

James Lake (Chair)
Associate Professor of Economics
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University of Tennessee
Email: jlake7@utk.edu

Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
University of Tennessee
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Matthew Van Essen
Lena M. and Clarence A. Garvin, Sr. Professor and Dr. Joe Johnson Faculty Fellow
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